Standard Operating Procedure for Resolving Critical Hardware/Environmental Faults on Ericsson PE HEB-PE-12 at Hebbal Edge

Node: HEB-PE-12 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_1715

Description

This SOP outlines the procedures for isolating and resolving critical hardware/environmental faults on the Ericsson PE HEB-PE-12 at the Hebbal Edge site. The PE role is responsible for processing and forwarding packets, and the Ericsson equipment is designed to operate in a high-availability environment. However, in the event of an ASIC memory parity error or line card crash, the PE may experience a loss of redundancy, leading to a critical fault. This fault is triggered when the Alarm Code ALARM_1715 is raised, indicating a critical hardware/environmental issue. Understanding the underlying protocol and hardware mechanics is essential to resolving this issue. The PE uses a combination of ASICs and line cards to process and forward packets, and any failures in these components can lead to a loss of redundancy and a critical fault.

Pre-requisite Checks

- The Hebbal Edge site must be accessible and the remote hands must be dispatched to the site to facilitate physical checks and repairs.

- The redundant link status must be verified to ensure that the link is operational and not contributing to the fault.

- The Ericsson equipment must be powered off to prevent any further damage or instability.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show tech-support | include ALARM_1715, Expected Output: ALARM_1715: Critical hardware/environmental fault detected, Description: This command is used to verify that the ALARM_1715 is raised and to obtain a detailed output of the fault.

Command: show system redundancy | include PE-12, Expected Output: PE-12: Redundant link status: UP, Description: This command is used to verify the redundant link status and ensure that it is operational.
```

```
Command: show system hardware | include PE-12, Expected Output: PE-12: ASIC memory parity error detected, Description: This command is used to verify the ASIC memory parity error and to obtain a detailed output of the fault.
```

Resolution Steps

Step 1: Step: Power off the Ericsson equipment, Command: system power-off, Description: This step is used to prevent any further damage or instability to the equipment.

Step 2: Step: Replace the faulty ASIC or line card, Command: system replace-part PE-12 ASIC/memory-parity-error, Description: This step is used to replace the faulty ASIC or line card and restore redundancy to the system.

Step 3: Step: Verify the redundant link status, Command: show system redundancy | include PE-12, Description: This step is used to verify that the redundant link status is operational and that the system is stable.

Post-Check Verification

```
Command: show system redundancy | include PE-12, Expected Output: PE-12: Redundant link status: UP, Description: This command is used to verify that the redundant link status is operational and that the system is stable.

Command: show system hardware | include PE-12, Expected Output: PE-12: ASIC memory parity error not detected, Description: This command is used to verify that the ASIC memory parity error is not detected and that the system is stable.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: system restore config, Description: This step is used to restore the original configuration and revert any changes made during the resolution process.

- Step: Power on the Ericsson equipment, Command: system power-on, Description: This step is used to power on the Ericsson equipment and restore normal operation.